

Course Title: Artificial Intelligence and Machine Learning

Semester: V

Term: ODD

Course Code: 24ITPCC502B

Teaching Scheme

Evaluation Scheme

Contact Hrs.			Credit Allotted			Total Credit	IAE 1	IAE 2	ESE	CA	Oral/Pract/Tut.	Total
Th	Tu	Pr	Th	Tu	Pr							
3	--	2	3	--	1	4	20	20	60	25	--	125

IAE: Internal Assessment Examination

ESE: End Semester Examination

CA: Continuous Assessment, TW: Term Work

Course Objectives:

The course aims:

- 1.To introduce the foundational principles of Artificial Intelligence, including intelligent agents and their environments.
- 2.To enable students to formulate real-world problems in a structured format and apply various search techniques for effective problem-solving.
- 3.To impart knowledge of logic-based knowledge representation and reasoning strategies for intelligent decision-making and planning.
- 4.To provide a comprehensive understanding of supervised learning models, focusing on classification and regression for predictive analytics.
- 5.To explore unsupervised learning techniques such as clustering and dimensionality reduction for discovering hidden patterns in data.
- 6.To develop a basic understanding of deep learning concepts through neural network models, training strategies, and estimation techniques.

Course Outcomes:

At the end of the course students will be able to:

1. Develop a basic understanding of the building blocks of AI as presented in terms of intelligent agents
2. Apply an appropriate problem-solving method and knowledge-representation scheme.
3. Develop an ability to analyze and formalize the problem (as a state space, graph, etc.). They will be able to evaluate and select the appropriate search method
4. Integrate supervised machine learning algorithms in a range of real-world applications.
5. Integrate unsupervised machine learning algorithms in a range of real-world applications.
6. Formulate Deep Learning Algorithms.

Module	Content	Hours	COs
0	Prerequisite: Data Structures and Algorithms, Statistics and Probability, Differentiation and Integration	0	
I	Introduction to AI Introduction: Introduction to AI, AI techniques, Problem Formulation. Intelligent Agents: Structure of Intelligent agents, Types of Agents, Agent Environments PEAS representation for an Agent. Self-Learning Topics : Identify application areas of AI	6	CO1
II	Search Techniques Uninformed Search Techniques: Uniform cost search, Depth Limited Search, Iterative Deepening, Bidirectional search. Informed Search Methods: Heuristic functions, Best First Search, A*, Hill Climbing, Simulated Annealing. Constraint Satisfaction Problem Solving: Cryptarithmic Problem, Water Jug, Graph Coloring. Adversarial Search: Game Playing, Min-Max Search, Alpha Beta Pruning. Comparing Different Techniques. Self-Learning Topics : IDA*, SMA*	9	CO2
III	Knowledge Representation using First Order Logic Knowledge and Reasoning: A Knowledge Based Agent, WUMPUS WORLD Environment, Propositional Logic, First Order Predicate Logic, Forward and Backward Chaining, Resolution. Planning as an application of a knowledge based agent. Concepts of Partial Order planning, Hierarchical Planning and Conditional Planning. Self-Learning Topics: Representing real world problems as planning problems	8	CO3
IV	Supervised Learning: Introduction to Supervised Learning, Examples of various datasets, Classification and Regression model:- Linear Regression, Logistic Regression, Polynomial Regression, Decision Tree, Random Forest, Naïve Bayes, SVM Self-Learning Topics: Real world case studies on Supervised Learning Algorithms.	8	CO4
V	Unsupervised Learning: Introduction to Unsupervised Learning, Example of various datasets, Clustering algorithm: - K-Means, Dimensionality reduction: - Principal Component Analysis. Self-Learning Topics: Real world case studies on Unsupervised Learning Algorithms.	7	CO5
VI	Neural Nets: Introduction, Early Models, Perceptron Learning, Backpropagation, Initialization, Training & Validation, Parameter Estimation - MLE, MAP, Bayesian Estimation. Self-Learning Topics: Real world case studies on Deep Learning Algorithms.	7	CO6
	Total	45	

Exp. No.	List of Experiments	CO
1	Explore following python libraries: numpy, pandas, sklearn, matplotlib	CO1
2	To Build an Intelligent agent	CO1
3	Implement Uninformed search techniques	CO2
4	Implement Informed Search Methods	CO2
5	Implement examples of Predicate Logic, for forward and backward reasoning and resolution using prolog	CO3
6	Write a program to implement Random Forest to classify the iris data set. Print both correct and wrong predictions.	CO4
7	Write a program to implement SVM algorithm to classify the iris dataset	CO4
8	Implement K means Clustering algorithm on dataset. Visualize the results for the same.	CO5
9	Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets.	CO6
10	Implementation of Mini project using case studies of Artificial Intelligence and Machine learning	CO1-CO6

Evaluation and Assessment Scheme :

A. Internal Assessment Examination:

Assessment consists of two class tests, each 20 marks. The IAE 1 will cover any three Course Outcomes (COs) and IAE 2 will cover the remaining three Course Outcomes (COs). Each test will have a duration of one hour.

B. End Semester Theory Examination:

The End Semester exam of 60 Marks will be conducted based on the entire syllabus.

C. Continuous Assessment (CA):

Term work should consist of 8 - 10 experiments / tutorials (whichever is applicable)

Journal must include at least 02 assignments based on different pedagogy in group wise manner

The final certification and acceptance of term work ensures the satisfactory performance of laboratory work, attendance and minimum passing marks in term work.

Experiments/ Tutorials: 10 marks

Attendance (Theory & Practical): 05 marks

Teacher Assessment Examination (TAE): 10 Marks

List of Teacher Assessment Examination (TAE):

1. Assignment
2. Case Study
3. Debate
4. Solution for Social Problems
5. Field Visit

6. Group Project
7. Flip Classroom
8. Topic Review
9. Quiz
10. Mind Mapping
11. Any other

Note: Number of activities to be conducted under TAE would be as per the subject need.

Reference Books:

1. Elaine Rich And Kevin Knight - "Artificial Intelligence", Tata Mcgraw-Hill
2. Saikat Dull, S. Chandramauli, Das - "Machine Learning", Pearson
3. Michael Nielsen "Neural Networks and Deep Learning", Determination Press.
4. Li Deng and Dong Yu - "Deep Learning: Methods and Applications", Now Publishers.

Text books:

1. Deepak Khemani, A First Course in Artificial Intelligence, McGraw Hill Publication
2. Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach, 2nd Edition, Pearson Education.
3. Ethem Alpaydm, "Introduction to Machine Learning", MIT Press

Useful Links:

1. <https://www.coursera.org/learn/machine-learning/home/info>
2. Sudeshna Sarkar, "Introduction to Machine Learning", IIT Kharagpur
3. <https://nptel.ac.in/courses/106105152/1>
4. <https://nptel.ac.in/courses/106106139/1>
5. <https://www.geeksforgeeks.org/machine-learning/>
6. https://www.tutorialspoint.com/machine_learning_with_python/index.htm